

Machine Learning & NumPy Lab Programs

1. NumPy Average Sale Price

Aim: Find average sale price of houses with >4 bedrooms.

Algorithm:

1. Import NumPy
2. Create array
3. Filter bedrooms>4
4. Find mean sale price
5. Display result

Program:

```
import numpy as np
data=np.array([[3,1400,7500000],[5,2200,12500000],[4,1800,9800000],[6,3000,18500000],
[2,1100,5200000]])
avg=np.mean(data[data[:,0]>4][:,2])
print("Average Sale Price:",avg)
```

Sample Output:

Average Sale Price: 15500000.0

Result:

Average sale price calculated successfully.

2. Logistic Regression - Spam

Aim: Classify email as spam/non-spam.

Algorithm:

6. Import libraries
7. Create dataset
8. Split X,y
9. Train model
10. Predict new email

Program:

```
from sklearn.linear_model import LogisticRegression
import pandas as pd
df=pd.DataFrame({'Num_Links':[1,6,0,4,2],'Num_Caps_Words':[3,15,1,10,5],'Email_Length':[120,300,80,250,160],'Spam':[0,1,0,1,0]})
X=df[['Num_Links','Num_Caps_Words','Email_Length'];y=df['Spam']
```

```
m=LogisticRegression().fit(X,y)
print(m.predict([[5,12,280]]))
```

Sample Output:

[1] (Spam)

Result:

Spam predicted successfully.

3. Logistic Regression

Aim: Same as Scenario 2.

Algorithm:

11. Import

12. Train

13. Predict

Program:

Same program as Scenario 2

Sample Output:

[1] (Spam)

Result:

Prediction completed.

4. Linear Regression

Aim: Predict car resale price.

Algorithm:

14. Import

15. Create dataset

16. Train model

17. Predict price

Program:

```
from sklearn.linear_model import LinearRegression
import pandas as pd
df=pd.DataFrame({'Age':[2,5,3,7,1],'Kms':[25000,60000,40000,90000,15000],'CC':[1200,1500,1300,1800,1000],'Price':[650000,420000,520000,300000,720000]})
X=df[['Age','Kms','CC'];y=df['Price']
m=LinearRegression().fit(X,y)
print(m.predict([[4,50000,1400]]))
```

Sample Output:

Predicted Price: 470000 (approx)

Result:

Resale price predicted successfully.

5. CART Decision Tree

Aim: Loan approval prediction.

Algorithm:

18. Import

19. Create dataset

20. Train Decision Tree

21. Predict applicant

Program:

```
from sklearn.tree import DecisionTreeClassifier
import pandas as pd
df=pd.DataFrame({'Income':[45000,30000,60000,28000,52000],'Credit':[720,650,780,620,710],
'Loans':[0,1,0,2,1], 'Approved':[1,0,1,0,1]})
X=df[['Income','Credit','Loans']];y=df['Approved']
m=DecisionTreeClassifier().fit(X,y)
print(m.predict([[50000,730,0]]))
```

Sample Output:

[1] (Approved)

Result:

Loan approval predicted successfully.